

Safety data sheet

Page: 1/13

Chemetall (now part of BASF Group) BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 2.1

Product: **GARDOBOND-ADDITIVE H 7004**

(ID no. 30705804/SDS_GEN_00/EN)

Date of print 11.03.2022

1. Identification

Product identifier

GARDOBOND-ADDITIVE H 7004

Recommended use: Additive for phosphating process.

Not recommended use: Uses other than recommended

Details of the supplier of the safety data sheet

Company:
Chemetall GmbH
Trakehner Straße, 3
60487, Frankfurt am Main
Germany
+49(0)69 7165-0
sds.global-chemetall@basf.com

Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Acute Tox. 4 (oral)
Eye Dam./Irrit. 2A
Aquatic Acute 1

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 2.1

Product: **GARDOBOND-ADDITIVE H 7004**

(ID no. 30705804/SDS_GEN_00/EN)

Date of print 11.03.2022

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System, UN (GHS)

Pictogram:



Signal Word:

Warning

Hazard Statement:

H319	Causes serious eye irritation.
H302	Harmful if swallowed.
H400	Very toxic to aquatic life.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P273	Avoid release to the environment.
P270	Do not eat, drink or smoke when using this product.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P330	Rinse mouth
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P301 + P312	IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell.
P391	Collect spillage.
P337 + P313	If eye irritation persists: Get medical attention.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.
------	---

Other hazards

According to UN GHS criteria

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 2.1

Product: **GARDOBOND-ADDITIVE H 7004**

(ID no. 30705804/SDS_GEN_00/EN)

Date of print 11.03.2022

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

Water, inorganic salts

Hazardous ingredients (GHS)

According to UN GHS criteria

Sodium nitrite

Content (W/W): $\geq 30\%$ - $< 50\%$

CAS Number: 7632-00-0

EC-Number: 231-555-9

Ox. Sol. 2

Acute Tox. 3 (oral)

Eye Dam./Irrit. 2A

Aquatic Acute 1

M-factor acute: 1

H272, H319, H301, H400

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Remove affected person from danger area. Keep warm, calm and covered up. Immediately remove contaminated clothing. Never give anything by mouth to an unconscious person. In case of intoxication, call a poison control center or physician for treatment advice, taking the packaging or the label of the product. Symptoms of poisoning may occur even after several hours, continue medical observation for at least 48 hours after the accident.

If inhaled:

Remove the affected individual into fresh air and keep the person calm. If symptoms persist, seek medical advice. If breathing is irregular or stopped, administer artificial respiration.

On skin contact:

Flush with copious amounts of water for at least 15 minutes. Remove contaminated clothing immediately and clean before re-use or dispose it if necessary. Immediate medical attention required.

On contact with eyes:

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 2.1

Product: **GARDOBOND-ADDITIVE H 7004**

(ID no. 30705804/SDS_GEN_00/EN)

Date of print 11.03.2022

Remove contact lenses, if present. Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist. Immediate medical attention required.

On ingestion:

Summon medical aid without delay. Do not induce vomiting due to aspiration hazard. Rinse mouth immediately with water. Keep at rest.

Most important symptoms and effects, both acute and delayed

Symptoms: Eye irritation, Symptoms can appear later., Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

Hazards: Danger of methaemoglobin formation.

Indication of any immediate medical attention and special treatment needed

Treatment: Symptomatic treatment (decontamination, vital functions).

Antidote: No known specific antidote.

5. Fire-Fighting Measures**Extinguishing media**

Suitable extinguishing media:

carbon dioxide, alcohol-resistant foam, dry powder, water spray

Unsuitable extinguishing media for safety reasons:

water jet

Special hazards arising from the substance or mixture

nitrogen oxides

When dry, the substance/product is oxidising. Hazardous decomposition products formed under fire conditions.

Advice for fire-fighters

Special protective equipment:

Appropriate breathing apparatus may be required.

Further information:

Cool closed containers in the vicinity of the source of fire. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. Product itself is non-combustible; fire extinguishing method of surrounding areas must be considered. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems.

6. Accidental Release Measures**Personal precautions, protective equipment and emergency procedures**

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 2.1

Product: **GARDOBOND-ADDITIVE H 7004**

(ID no. 30705804/SDS_GEN_00/EN)

Date of print 11.03.2022

Avoid breathing vapours. For non-emergency personnel: Use personal protective clothing. Ensure adequate ventilation. Keep away from sources of ignition. For emergency responders: Advice on product handling can be found in sections 7 and 8 of this safety data sheet. Information regarding personal protective measures, see section 8.

Environmental precautions

Do not allow to enter drains or waterways. Do not discharge into the subsoil/soil. If the product enters drains or sewers, the local water company should be contacted immediately; in the case of contamination of streams, rivers or lakes, the Environment Agency.

Methods and material for containment and cleaning up

Contain and collect spillage with non-combustible absorbent materials, e.g. sand, earth, vermiculite, diatomaceous earth and place in a suitable container for disposal according with the waste regulations (see section 13). Clean preferably with a detergent; avoid the use of solvents. Ensure adequate ventilation.

7. Handling and Storage**Precautions for safe handling**

Provide good ventilation of working area (local exhaust ventilation if necessary). Do not return residues to the storage containers. Smoking, eating and drinking are forbidden in application area. For personal protection see section 8. Comply with the health and safety at work laws. Avoid inhalation of vapour and spray mist. The workplace should be equipped with an emergency shower and eye-rinsing facility. Avoid contact with the skin, eyes and clothing. Handle in accordance with good industrial hygiene and safety practice.

Protection against fire and explosion:

Avoid all sources of ignition: heat, sparks, open flame. The relevant fire protection measures should be noted. Dried product promotes spreading of fire.

Conditions for safe storage, including any incompatibilities

Segregate from ammonium salts. Segregate from organic substances. Keep away from amines, oxidizing agents, from strongly alkaline and strongly acid materials. Segregate from combustible materials. Segregate from reducing agents.

Suitable materials for containers: Polypropylene (PP), Stainless steel 1.4301 (V2), Polyethyleneterephthalate (PET), Low density polyethylene (LDPE), High density polyethylene (HDPE)

Further information on storage conditions: Keep container dry. Keep in a cool, well-ventilated place. Avoid direct sunlight. Close containers carefully once opened and store them upright in order to prevent any leakage. No smoking. No admission for unauthorised personnel. Always keep in containers of same material as the original one. Observe label precautions.

Storage stability:

Storage temperature: -14 - 40 °C

Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 2.1

Product: **GARDOBOND-ADDITIVE H 7004**

(ID no. 30705804/SDS_GEN_00/EN)

Date of print 11.03.2022

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

No occupational exposure limits known.

Exposure controls

Personal protective equipment

Respiratory protection:

Wear respiratory protection if ventilation is inadequate. (Gas filter EN 14387 B)

Hand protection:

Chemical resistant protective gloves (EN ISO 374-1)

nitrile rubber (NBR) - 0.4 mm coating thickness

butyl rubber gloves - material thickness: 0.5 mm

Performance level 6, corresponding to a breakthrough time of >480 min according to EN ISO 374-1

The protection glove should be tested for its specific suitability (e.g. mechanical strength, product compatibility, anti-static properties).

The gloves should be replaced immediately in case of damage or signs of wear. It is recommended to use preventative skin protection (skin cream).

Eye protection:

Tightly fitting safety goggles (splash goggles) (e.g. EN 166)

Body protection:

Chemical resistant protective clothing according to DIN EN 13034 (Type 6)

General safety and hygiene measures

Do not breathe vapour/spray. Eye wash fountains and safety showers must be easily accessible.

Avoid contact with the skin, eyes and clothing. Handle in accordance with good industrial hygiene and safety practice. Ensure adequate ventilation. This can be achieved by the use of local exhaust ventilation and good general extraction. If these are not sufficient to maintain concentrations at the workplace below the occupational exposure limits, appropriate certified respirators must be worn.

Remove contaminated clothing immediately and dispose of safely. Hands and/or face should be washed before breaks and at the end of the shift. Keep separated from food stuffs and feed stocks.

9. Physical and Chemical Properties

Information on basic physical and chemical properties

Form:	liquid
Colour:	light yellow
Odour:	odourless

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 2.1

Product: **GARDOBOND-ADDITIVE H 7004**

(ID no. 30705804/SDS_GEN_00/EN)

Date of print 11.03.2022

pH value:	8,6 - 9,6 (20 °C) (undiluted)
Melting point:	< -14 °C
onset of boiling:	not determined
Flash point:	not applicable
Flammability:	not applicable
Lower explosion limit:	not determined
Ignition temperature:	not determined
Vapour pressure:	(20 °C) not determined
Density:	(50 °C) not determined 1,300 g/cm ³ (20 °C)
Solubility in water:	fully soluble
Self ignition:	not self-igniting
Viscosity, kinematic:	9,8 mm ² /s (20 °C)
Explosion hazard:	(40 °C) not determined
Fire promoting properties:	not explosive not fire-propagating

Other information

Self heating ability: It is not a substance capable of spontaneous heating.

Miscibility with water:

miscible

Flow time: 35 s

(DIN EN ISO 2431; 3 mm)

10. Stability and Reactivity**Reactivity**

No hazardous reactions if stored and handled as prescribed/indicated.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 2.1

Product: **GARDOBOND-ADDITIVE H 7004**

(ID no. 30705804/SDS_GEN_00/EN)

Date of print 11.03.2022

Possibility of hazardous reactions

Contact with acids may liberate toxic gas. Dried product promotes spreading of fire.

Conditions to avoid

Avoid direct sunlight. Avoid drying-out.

Incompatible materials

Substances to avoid:

flammable, oxidizable substances, organic substances, ammonium salts, Keep away from highly acidic or alkaline substances, amines and oxidants in order to prevent exothermal reactions., reducing agents

Hazardous decomposition products

:

When exposed to high temperatures hazardous decomposition products such as carbon monoxide, carbon dioxide, smoke, oxides of nitrogen may be produced., No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information**Information on toxicological effects**Acute toxicity

Assessment of acute toxicity:

Of moderate toxicity after single ingestion.

Information on: Sodium nitrite

Experimental/calculated data:

LD50 rat (oral): 180 mg/kg

Irritation

Assessment of irritating effects:

Not irritating to the skin. Eye contact causes irritation.

Respiratory/Skin sensitization

Assessment of sensitization:

Based on available data, the classification criteria are not met.

Germ cell mutagenicity

Assessment of mutagenicity:

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 2.1

Product: **GARDOBOND-ADDITIVE H 7004**

(ID no. 30705804/SDS_GEN_00/EN)

Date of print 11.03.2022

Based on available data, the classification criteria are not met.

Carcinogenicity

Assessment of carcinogenicity:

Based on available data, the classification criteria are not met.

Reproductive toxicity

Assessment of reproduction toxicity:

Based on available data, the classification criteria are not met.

Developmental toxicity

Assessment of teratogenicity:

Based on available data, the classification criteria are not met.

Specific target organ toxicity (single exposure)

Assessment of STOT single:

Based on available data, the classification criteria are not met.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

Based on available data, the classification criteria are not met.

Aspiration hazard

No aspiration hazard expected.

Other relevant toxicity information

Inhalation of nitrous gas (e.g. after fires) can cause lung oedema.

Inhalation of nitrous gas (e.g. after fires) can cause lung oedema. Nausea, unconsciousness, haematurie (blood in the urine), shortness of breath or circulatory collapse are possible.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:

Very toxic to aquatic life. There are no test results available for this product. Do not allow to enter drains or waterways.

Persistence and degradability

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 2.1

Product: **GARDOBOND-ADDITIVE H 7004**

(ID no. 30705804/SDS_GEN_00/EN)

Date of print 11.03.2022

Assessment biodegradation and elimination (H₂O):

Biological degradability of hazardous substances mentioned in section 3:

*Information on: Sodium nitrite**Elimination information:*not applicable
-----**Bioaccumulative potential**

Bioaccumulation potential:

No data available.

Mobility in soil

Assessment transport between environmental compartments:

Adsorption in soil: No data available.

Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

13. Disposal Considerations**Waste treatment methods**

Observe national and local legal requirements.

No disposal via sewage or waste water systems.

Contaminated packaging:

Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

14. Transport Information**Land transport**

ADR

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 2.1

Product: **GARDOBOND-ADDITIVE H 7004**

(ID no. 30705804/SDS_GEN_00/EN)

Date of print 11.03.2022

UN number UN3082
UN proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID,
N.O.S. (contains SODIUM NITRITE)
Transport hazard class(es): 9, EHS
Packing group: III
Environmental hazards: yes
Special precautions for user:

RID

UN number UN3082
UN proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID,
N.O.S. (contains SODIUM NITRITE)
Transport hazard class(es): 9, EHS
Packing group: III
Environmental hazards: yes
Special precautions for user: None known

Inland waterway transport**ADN**

UN number UN3082
UN proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID,
N.O.S. (contains SODIUM NITRITE)
Transport hazard class(es): 9, EHS
Packing group: III
Environmental hazards: yes
Special precautions for user: None known

Transport in inland waterway vessel

Not evaluated

Sea transport**IMDG**

UN number: UN 3082
UN proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID,
N.O.S. (contains SODIUM NITRITE)
Transport hazard class(es): 9, EHS
Packing group: III
Environmental hazards: yes
Marine pollutant: YES
Special precautions for user: None known

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 2.1

Product: **GARDOBOND-ADDITIVE H 7004**

(ID no. 30705804/SDS_GEN_00/EN)

Date of print 11.03.2022

Air transport

IATA/ICAO

UN number:	UN 3082
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains SODIUM NITRITE)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

Transport in bulk according to Annex II of MARPOL and the IBC Code

Regulation:	Not evaluated
Shipment approved:	Not evaluated
Pollution name:	Not evaluated
Pollution category:	Not evaluated
Ship Type:	Not evaluated

15. Regulatory Information**Safety, health and environmental regulations/legislation specific for the substance or mixture**

Not applicable

16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Acute Tox.	Acute toxicity
Eye Dam./Irrit.	Serious eye damage/eye irritation
Aquatic Acute	Hazardous to the aquatic environment - acute
Ox. Sol.	Oxidising solids
H272	May intensify fire; oxidizer.
H319	Causes serious eye irritation.
H301	Toxic if swallowed.
H400	Very toxic to aquatic life.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 26.08.2021

Version: 2.1

Product: **GARDOBOND-ADDITIVE H 7004**

(ID no. 30705804/SDS_GEN_00/EN)

Date of print 11.03.2022

responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.